

NATIONAL INSTITUTE FOR HEALTH AND CARE EXCELLENCE

NICE guidelines

Equality impact assessment

Osteoporosis: risk assessment

The impact on equality has been assessed during guidance development according to the principles of the NICE equality policy.

The initial scope and EIA for osteoporosis also covered a broader range of topics. The guideline will now be developed in two parts: part 1 covers risk assessment, vertebral fracture identification and treatment criteria; part 2 covers treatment (pharmacological, exercise, calcium and vitamin D), follow up for those on treatment and treatment pauses. This EIA only covers issues related to part 1.

1.0 Checking for updates and scope: before scope consultation (to be completed by the Developer and submitted with the draft scope for consultation)

1.1 Is the proposed primary focus of the guideline a population with a specific communication or engagement need, related to disability, age, or other equality consideration? Y/N

If so, what is it and what action might be taken by NICE or the developer to meet this need? (For example, adjustments to committee processes, additional forms of consultation.)

The proposed focus of the guideline is not a population with specific communication or engagement need.

No requirement for specific adjustment to processes has been identified.

1.2 Have any potential equality issues been identified during the check for an update or during development of the draft scope, and, if so, what are they?

- **Age –**

- Osteoporosis is more prevalent in older people and is still dismissed as ‘an inevitable part of the aging process’ resulting in under-diagnosis and under-treatment in the elderly. Conversely, diagnosis is commonly delayed or not considered in young people presenting with osteoporosis because of the well-recognised link to aging.
- Very old age is also associated with low level of physical activity which increases the risk of fracture.

- **Disability –**

- There could be difficulty adhering to the complex instructions for taking bisphosphonates for some groups (dementia, learning disabilities, those unable to remain upright for the specified time period and patients in whom oral bisphosphonates might be contraindicated such as those with oesophageal stricture). There is also evidence showing higher risk of osteoporosis in several of these groups e.g. learning disability.
- People with disabilities are associated with low levels of physical activity which is associated with increased risk of fracture.
- A recently published study found that fracture rates are considerably higher in people with intellectual disability than those without. Current osteoporosis guidelines do not identify people with intellectual disability as at risk of fracture, thereby missing a potential opportunity for fracture prevention.

- **Gender reassignment** - Hormone blockers used in gender reassignment are associated with bone loss. This can be counteracted by use of hormone replacement. Transgender men’s risk of osteoporosis may be overlooked if they transitioned at a late age.

- **Pregnancy and maternity** – Pregnancy and lactation associated osteoporosis tend to be diagnosed very late and therefore may be overlooked.

- **Race** - Women of Asian or European family origin are at a higher risk for osteoporosis than those of other ethnicities. Although osteoporosis is less common in other races, it occurs across the board and could be overlooked in other ethnicities. It is particularly important to assess risk in the presence of fragility fracture and other risk factors such as steroid use.

- **Religion or belief** - none

- **Sex**
 - Women are at higher risk than men of developing osteoporosis, and particularly women who are older and post-menopausal. By contrast, osteoporosis is under diagnosed in men and men with low bone density t-scores have a high risk of fracture but are often overlooked.
 - Some medications are only licenced for treatment in post-menopausal women (e.g. romosozumab, ibandronic acid, raloxifene). Therefore, younger women and men with osteoporosis may be disadvantaged as there are less treatments available. Some medications are only licenced for treatment in post-menopausal women and in men (e.g. denosumab, risedronate, teriparatide, zoledronic acid, strontium ranelate) therefore women of child bearing age may be disadvantaged. I
- **Sexual orientation** - none
- **Socio-economic factors** - there is increasing evidence linking poor bone health with low socioeconomic status. Low socioeconomic groups may have a higher prevalence of other predisposing factors including disability.
- **Other definable characteristics:**
 - refugees, asylum seekers and people who are homeless are at risk of not being able to follow up their treatment because they may not have a fixed address.
 - prisoners and young offenders – at risk of not being able to follow up their treatment.

1.3 What is the preliminary view on the extent to which these potential equality issues need addressing by the Committee?

When making protocols the committee will be asked whether any of these factors need to be analysed as subgroups.

Groups in which medications are not licenced could have recommendations made with a note they are off label.

It is noted that the groups identified above are important to consider when making recommendations for the guideline. All will be included within the population covered by the reviews. Where appropriate they will be considered as subgroups when the protocols for specific review questions are set.

1.3 What is the preliminary view on the extent to which these potential equality issues need addressing by the Committee?

This guideline will cross refer to guidelines on Patient experience (CG139), Shared decision making (NG197) and Decision-making and mental capacity (NG108) that address communication issues.

Problems with access to healthcare for people from lower socio-economic groups, refugees, asylum seekers and people who are homeless relate to a wider issue than can be dealt with in this guideline.

Completed by Developer: Carlos Sharpin

Date: 12th October 2022

Approved by NICE quality assurance lead: Simon Ellis

Date: 14 October 2022

2.0 Checking for updates and scope: after consultation (to be completed by the Developer and submitted with the revised scope)

2.1 Have any potential equality issues been identified during consultation, and, if so, what are they?

- Age – the following additional points were identified during consultation
 - It was noted that there is a significant group of patients who present to medical hospitals wards and fall outside the remit of the Fracture Liaison Service (FLS). These include older people admitted predominately with vertebral and pubic rami fractures, who are high risk of further fragility fractures and special recommendations should be considered for these patients.
 - It was also noted that treatment compliance (the way the medication is taken) and treatment persistence (continuing with the medication long-term) are both important for older people, particularly those with swallowing problems, polypharmacy and those with cognitive issues.
- Disability
 - There was a suggestion that the information and support needs for people with cognitive impairment, a learning disability, autism or any other disability needs a separate review question.
- Gender reassignment – no additional issues raised during consultation
- Pregnancy and maternity – no additional issues raised during consultation
- Race
 - There was a suggestion that women from an Asian background had a problem accessing care related to a language barrier. This group was more likely to not be able to speak English and therefore would need to attend with one of their children or a partner.
- Religion or belief – no additional issues raised during consultation
- Sex – the only issue raised during consultation is listed in section on race
- Sexual orientation – no issues raised during consultation
- Socio-economic factors – no issues raised during consultation
- Other definable characteristics:
 - Difficulties in access was raised in relation to people who live far away from scanners. This is likely to be exacerbated for people who do not have the money to travel.

2.2 Have any changes to the scope been made as a result of consultation to highlight potential equality issues?

An additional question was added to the scope on the information and support needs of adults who are at risk of fragility fracture or who have osteoporosis and cognitive impairment, a learning disability or autism. The guideline will also cross refer to the NICE guideline on Decision making and mental capacity (<https://www.nice.org.uk/guidance/NG108>).

No other changes were made.

With regard to the issue relating people falling outside the remit of the Fracture Liaison Service the scope aims to cover all settings where NHS-funded care or social care is provided or commissioned. The committee will consider making separate recommendations for older people where the evidence and their experience permits.

The issue relating to treatment compliance and persistence will be addressed in the scope as the outcome treatment adherence in the evidence reviews. It will also be considered as part of the risk reassessment as this is believed to help improve adherence. The committee will consider making specific recommendations if the evidence allows. General recommendations related to supporting adherence are not included in this guideline as they are covered by the NICE guideline on Medicines adherence (<https://www.nice.org.uk/guidance/cg76>).

Generic recommendations that may help alleviate the language barrier are covered in NICE's guideline on Patient experience in adult NHS services (<https://www.nice.org.uk/guidance/CG138>) and the NICE guideline on Shared decision making (<https://www.nice.org.uk/guidance/ng197>). Encouraging people to access medical services is beyond the scope of this guideline.

Difficulties in accessing treatment because of distance to scanners will be considered indirectly as part of the resource impact of recommendations. However, other aspects of service delivery are beyond the scope of this guideline.

2.3 Have any of the changes made led to a change in the primary focus of the guideline which would require consideration of a specific communication or engagement need, related to disability, age, or other equality consideration?

If so, what is it and what action might be taken by NICE or the developer to meet this need? (For example, adjustments to committee processes, additional forms of consultation)
No

Updated by Developer: Carlos Sharpin

Date: 15/02/2023

Approved by NICE quality assurance lead: Simon Ellis

Date: 16/02/2023

3.0 Guideline development: before consultation (to be completed by the Developer before consultation on the draft guideline)

3.1 Have the potential equality issues identified during the scoping process been addressed by the Committee, and, if so, how?

- Age

To take into account that the risk of fracture increases with age the guideline has made different recommendations for risk assessment by age group.

Vertebral fragility fractures were also highlighted as often being missed and particularly in older people. The guideline aimed to cover this with 3 review questions related to vertebral fragility fracture identification. This resulted in a research recommendation for the Vfrac risk assessment tool for 1 review; recommendations on when to consider vertebral fracture assessment (VFA) when doing a dual-energy X-ray absorptiometry (DXA); and a link to the [NICE health technology evaluation on artificial intelligence \(AI\) technologies to aid opportunistic detection of vertebral fragility fractures: early value assessment](#).

- Disability

The committee were aware of a new risk assessment tool for people with learning disabilities. However, at the time of finalising the draft recommendations a peer reviewed article was not available to include the evidence in the guideline. The articles relating to the clinical and cost effectiveness of this tool are waiting peer reviewed publication.

- Gender reassignment

Stakeholder consultation on the EIA noted that hormone blockers used in gender reassignment are associated with bone loss and that transgender men's risk of osteoporosis may be overlooked if they transitioned at a late age. Taking medicines that are associated with fracture risk is included as a risk factor for fragility fracture. The committee did not make specific recommendations for trans and non-binary people because the information available at the time of development for these groups of people was too limited.

- Pregnancy and maternity

A [surveillance review](#) in 2024 found no new evidence to suggest that pregnancy and postpartum is a risk factor for osteoporosis, as pregnancy associated osteoporosis is an ultra-rare condition. The conclusion was not to include it as a specific group within the guideline. The committee agreed that pregnancy associated osteoporosis would need specialist management.

- Race

No evidence was available to suggest different recommendations were needed in the guideline. Recommendations to address issues related to language

3.1 Have the potential equality issues identified during the scoping process been addressed by the Committee, and, if so, how?

barriers are covered in the [NICE guideline on patient experience in adult NHS services](#).

- Religion or belief

No issues identified.

- Sex

To take into account that the risk of fracture is higher for postmenopausal women than men of a similar age, the guideline has made recommendations for risk assessment by age group.

- Sexual orientation

No issues identified

- Socio-economic factors

No evidence was available to suggest specific recommendations were needed for this group.

- Other definable characteristics:

Refugees, asylum seekers and people who are homeless are at risk of not being able to follow up their treatment because they may not have a fixed address. While access issues for DXA in general have not been dealt with in this guideline people in this group are also covered by the recommendation about what to do when DXA is not always technically feasible or tolerated. This recommends that a shared decision with the person should be made about the appropriateness of treatment without the DXA results if the person meets the other criteria for treatment.

Prisoners and young offenders not being able to follow up their treatment is covered by recommendations in the [NICE guideline on physical health of people in prison](#).

3.2 Have any **other** potential equality issues (in addition to those identified during the scoping process) been identified, and, if so, how has the Committee addressed them?

3.3 Have the Committee's considerations of equality issues been described in the guideline for consultation, and, if so, where?
Recommendations have been made by age groups to reflect the increased risk of fragility fracture as people get older, and by sex to reflect the increased risk of fragility fracture in women, particularly following the menopause. This is discussed in the rationales and committee discussions in the evidence reviews.
3.4 Do the preliminary recommendations make it more difficult in practice for a specific group to access services compared with other groups? If so, what are the barriers to, or difficulties with, access for the specific group?
• Age and disability • DXA is not always technically feasible or tolerated (for example, in older people who are living with frailty or are housebound for whom DXA is not suitable).
3.5 Is there potential for the preliminary recommendations to have an adverse impact on people with disabilities because of something that is a consequence of the disability?
No issues were identified other than those covered in points 3.4 and 3.6.
3.6 Are there any recommendations or explanations that the Committee could make to remove or alleviate barriers to, or difficulties with, access to services identified in box 3.4, or otherwise fulfil NICE's obligation to advance equality?

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When DXA is not always technically feasible or tolerated the committee recommend that a shared decision with the person should be made about the appropriateness of treatment without the DXA results if the person meets the other criteria for treatment.

Completed by Developer: Carlos Sharpin

Date: 10/12/2025

Approved by NICE quality assurance lead: Sara Buckner

Date: 29/12/2025

4.0 Final guideline (to be completed by the Developer before GE consideration of final guideline)

4.1 Have any additional potential equality issues been raised during the consultation, and, if so, how has the Committee addressed them?

- Disability

Stakeholders identified that people with chronic serious mental illness may have a higher risk of fragility fractures due to a combination of risk factors. These may include long-term pharmacological treatment (for example antipsychotic, antidepressant medicines), lack of exercise, poor nutrition and higher tobacco smoking rate. People with methadone maintenance treatment were noted to have an increased risk of fragility fracture due to having a lower bone mineral density. These factors were noted but no changes were made to recommendations. A full clinical risk assessment is already recommended as part of assessing fragility fracture risk.

- Gender reassignment

Stakeholders suggested adding recommendations for trans and non-binary people despite a lack of evidence. The committee made consensus recommendations for trans and non-binary people for risk assessment and treatment criteria.

- Pregnancy and maternity

People with suspected rare bone disease associated with fragility fractures are recommended to seek advice from a specialist when assessing fragility fracture risk and for treatment options. Pregnancy-related osteoporosis has been added as an example of rare bone disease associated with fragility fractures in the recommendations.

- Race

The committee have recommended QFracture and FRAX risk tools that are validated in UK populations. It was noted that most of the population are white in the validation studies and therefore any potential variation in race is unlikely to be fully accounted for in the risk prediction tools. The committee have made a recommendation to complete a full risk assessment alongside the risk tools to consider all factors that are not accounted for in the tools such as race.

Stakeholder consultation comments highlighted ethnic variation in bone mineral density, osteoporosis prevalence and fracture outcomes that need to be considered. The committee made a recommendation that a full clinical assessment should be made alongside a risk tool to consider all relevant risk factors.

- Sex

4.1 Have any additional potential equality issues been raised during the consultation, and, if so, how has the Committee addressed them?

To take into account that the risk of fracture is higher for women who have experienced menopause than men of a similar age, recommendations were made for risk assessment by age group. Further clarification was added into the risk factors review that most women experience menopause between the ages of 45 and 55 years. Women with untreated early menopause or premature ovarian insufficiency were identified as having a greater risk of fragility fracture than women of a similar age and were therefore included in risk assessment recommendations for women who have not experienced menopause.

For clarification, in this document, gendered terms, such as women and men, are used widely and intended to correspond to biological sex terms female and male respectively unless otherwise explicitly stated (for example, trans and non-binary people). This reflects common historical clinical and epidemiological reporting conventions.

NICE uses person-centred language so recommendations use the terms 'women' and 'men' to reflect how people speak and how they wish to be referred to.

- Socio-economic factors

Stakeholders identified that socioeconomic deprivation is associated with lower bone mineral density and increased risk of osteoporosis in adulthood. This includes access to bone health services, including lower rates of DXA assessment, reduced initiation and persistence with anti-osteoporosis medication. These factors were added to the relevant committee discussion, but no changes were made to recommendations.

4.2 If the recommendations have changed after consultation, are there any recommendations that make it more difficult in practice for a specific group to access services compared with other groups? If so, what are the barriers to, or difficulties with, access for the specific group?

None

4.3 If the recommendations have changed after consultation, is there potential for the recommendations to have an adverse impact on people with disabilities because of something that is a consequence of the disability?

None

4.4 If the recommendations have changed after consultation, are there any recommendations or explanations that the Committee could make to remove or alleviate barriers to, or difficulties with, access to services identified in question 4.2, or otherwise fulfil NICE's obligations to advance equality?

None

4.5 Have the Committee's considerations of equality issues been described in the final guideline, and, if so, where?

The following considerations have been discussed in the 'other factors the committee took into account' section of the committee discussion in the following evidence reviews:

Evidence review A:

It was noted that socioeconomic deprivation is associated with lower bone mineral density and increased risk of osteoporosis in adulthood.

Evidence review B:

The committee acknowledged that there may be patient characteristics not yet accounted for on the electronic health records. For example, it is not currently possible to automatically identify someone who has a biological risk that differs from the gender of which they identify. The committee also noted that people who are not registered with a GP and do not have an electronic health record would not be picked up.

Evidence review C:

The committee noted that people with learning difficulties may find completing the forms difficult depending on the severity of their condition. However, the forms are generally completed by healthcare professionals rather than the individual themselves, but extended GP appointment time may be needed. People with learning disabilities or cognitive impairment may be less able to provide accurate answers in relation to risk factors that are not recorded in the medical records (for example, parental hip fracture).

Evidence review D:

Living in a remote area was another factor that could be a potential barrier to receiving care such as waiting time for DXA scans or possible need for

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transportation. Physical impairment was also considered as an example of additional challenges to accessing DXA scans.

Evidence review E:

DXA is not always technically feasible (for example, if there is a metal implant in both the hip and lumbar spine), or tolerated (for example in older people who are living with frailty or has a physical impairment that makes positioning on a scanner impossible). It was noted that some older people have good function and some younger people do not. The committee amended the recommendation, removing reference to 'older' and focusing on people living with frailty. It was also noted that Black women generally have higher bone mineral density and lower rates of fractures compared to White and Asian women, but they still experience osteoporosis. Living in remote areas was acknowledged as a potential barrier to receiving care, because of factors such as waiting time for DXA scans or need for transport. The committee also noted that some people with learning disabilities could struggle with the positioning needed for the DXA scan.

Stakeholders agreed with the recommendation to seek specialist advice for women who have not experienced menopause and men under 50 who have had a major osteoporotic fracture. However, they also noted that they could experience long waiting times for specialist advice in secondary care osteoporosis clinics and that this is likely to vary across the country. The committee discussed this and agreed that this recommendation is important because sustaining a fracture or losing bone mineral density at this age would be unusual and merit further consideration to establish the cause and appropriate treatment.

Evidence review F:

There could also be language barriers to completing the tool, although it has been translated into Urdu so it is possible this could be done for other languages.

Evidence review G:

The committee acknowledged that most evidence for the accuracy of DXA-based VFA was in women who have experienced menopause and that men and people with learning or physical disabilities are underrepresented.

Evidence review H:

The committee discussed that people with learning disabilities and cognitive impairment may need support when expecting them to self-refer for ongoing management.

Approved by Developer: Carlos Sharpin

Date: June 2026

Approved by NICE quality assurance lead: Sara Buckner

Date: June 2026